

Automated Web Data Extraction Pipeline

Python – Selenium – BeautifulSoup – NLTK

Chelsea's Project

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Project Overview

Python based project designed to automatically collect and extract textual information from the web. The goal of this project was to transform scattered online content into structured data that can be analyzed further. Using Selenium and BeautifulSoup, the system crawls search results, gathers relevant links, and retrieves the main text content from each webpage. The extracted data is then processed using Python and NLTK to perform basic text preprocessing and tokenization, making the raw web content easier to analyze. This project demonstrates how automated web scraping techniques can be used to build a lightweight data collection pipeline for transforming large volumes of unstructured web information into usable datasets.

Technology Stack

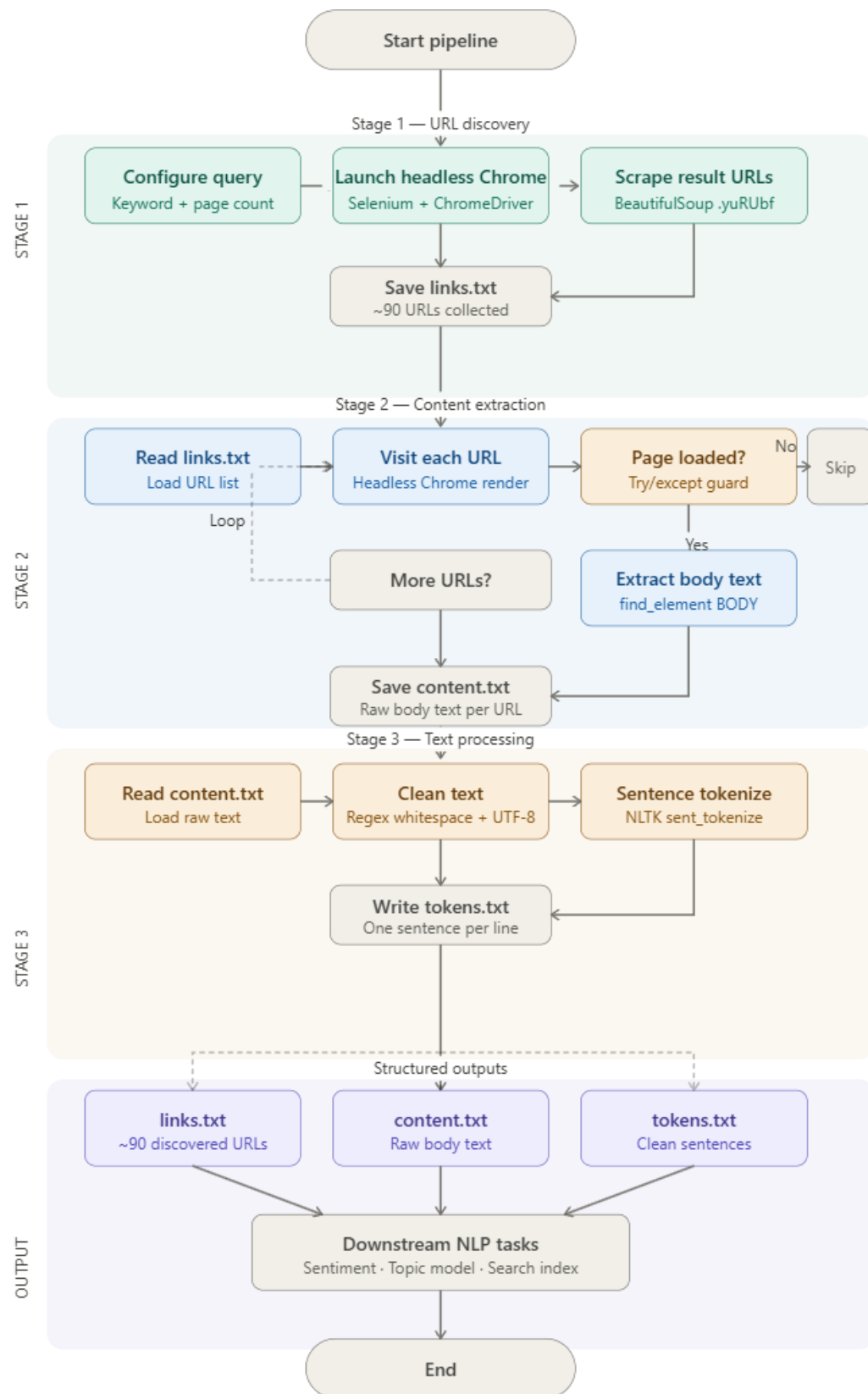
Libraries and tools

Library/tool	Purpose
Python	Core programming language
Selenium WebDriver	Headless browser automation & dynamic rendering
BeautifulSoup4	HTML parsing and element selection
Webdriver-manager	Automatic ChromeDriver version management
NLTK	Natural language text tokenization
argparse	CLI argument parsing for modular script execution

Why using Selenium and BeautifulSoup?

Many modern websites render content via JavaScript, making simple HTTP requests insufficient. Selenium drives a real Chrome browser (headlessly) to ensure fully rendered pages are captured before BeautifulSoup parses the resulting HTML. This combination handles both static and dynamic content reliably.

Roadmap Project



Implementation

Stage 1 – URL discovery (scraper.py)

This module queries Google Search for given keyword across multiple pages, parsing the result links using BeautifulSoup's CSS selector targeting Google's search result anchors.

```
from bs4 import BeautifulSoup
from selenium import webdriver
from webdriver_manager.chrome import ChromeDriverManager
chrome_options = webdriver.ChromeOptions()
chrome_options.add_argument("--headless")
driver = webdriver.Chrome(ChromeDriverManager().install(),
                           chrome_options=chrome_options)
#query to obtain links
query = 'Joko Widodo'
#links = ..
#specifi
n_pages = 10

for page in range(1, n_pages):
    url = "https://www.google.com/search?q=" + query + "&start=" +
        str((page-1) * 10)
    driver.get(url)
    soup = BeautifulSoup(driver.page_source, 'html.parser')
    #soup...

    search = soup.find_all('div', class_="yuRUbf")
    for h in search:
        #links..
        print(h.a.text)

    print(h.a.get('href'))
```

Stage 2 – Content Extraction (extractor.py)

This module reads a list of URLs from a file, visits each one with a headless Chrome driver, and captures the rendered body text. It accepts CLI arguments for flexible input/output configuration.

```
from selenium import webdriver
import sys, getopt
import argparse
from webdriver_manager.chrome import ChromeDriverManager
from selenium.webdriver.chrome.options import Options
from selenium.webdriver.chrome.service import Service
from selenium.webdriver.common.by import By

options = Options()
options.add_argument("--headless")
driver = webdriver.Chrome(service=Service(ChromeDriverManager().install()), options=options)
full_text=[]
```

```

def parse_args():
    parser = argparse.ArgumentParser()
    parser.add_argument('-i', '--infile', default='', help='input filename')
    parser.add_argument('-o', '--outfile', default='', help='output filename')
    return parser.parse_args()

def main():
    args = parse_args()
    outfile = args.outfile
    infile = args.infile

    with open(infile) as f:
        content = f.read().splitlines()
    f.close()

    f = open(outfile, "w")
    for u in content:
        driver.get(u)
        elems = driver.find_element(By.TAG_NAME, 'body').text.encode("utf-8")
        full_text.append(elems)
    print(full_text)

    f.write(str(full_text))

    #print(full_text)
    driver.close()
    f.close()

main()

```

Stage 3 – Text Processing (processor.py)

Raw extracted text is cleaned and tokenized at the sentence level using NLTK, producing structured output ready for downstream NLP tasks.

```

import nltk
from nltk.tokenize import sent_tokenize
import re

with open('dikitlagi.txt', 'r') as file:
    text = file.read() #membuka file dan menyimpan dalam text
    tokens = nltk.tokenize.sent_tokenize(text) #tokenisasi text
    array = text.split() #memecah text menjadi array

    with open('terakhir.txt', 'w') as file:
        file.write(text) #mencetak hasil

print(tokens, array)

```

Output

```
[{'b"PRESIDEN REPUBLIK INDONESIA\\nPresiden Joko Widodo\\nIr.', 'H. Joko Widodo\\nUnduh Foto Resmi\\nIr.', 'H. Joko Widodo adalah Presiden ke-7 Republik Indonesia yang mulai menjabat sejak 20 Oktober 2014.', 'Lahir di Surakarta, Jawa Tengah, pada 21 Juni 1961, Joko Widodo pertama kali terjun ke pemerintahan sebagai Wali Kota Surakarta (Solo) pada 28 Juli 2005 hingga 1 Oktober 2012.\\nSelepas itu, Joko Widodo menjabat sebagai Gubernur DKI Jakarta pada 15 Oktober 2012 sebelum terpilih sebagai Presiden Republik Indonesia pada Pemilihan Presiden (Pilpres) 2014.', 'Saat Pilpres tersebut Joko Widodo terpilih bersama pasangannya, Jusuf Kalla.\\nDalam Pilpres 2019, Joko Widodo kembali terpilih sebagai Presiden Republik Indonesia untuk masa jabatannya yang kedua.', 'Kali ini, Joko Widodo didampingi oleh Wakil Presiden K.H.', 'Ma\\xe2\\x80\\x99ruf Amin dan dilantik pada 20 Oktober 2019 untuk masa jabatan 2019 hingga 2024 mendatang.\\nPembangunan infrastruktur menjadi program prioritas di masa kepemimpinannya yang pertama.', 'Pembangunan yang dilakukan secara merata hingga ke daerah terluar Indonesia ini dilakukan untuk mengejar ketertinggalan Indonesia dalam sektor ini dibandingkan negara-negara lain.\\nProgram prioritas tersebut dibarengi dengan program berupa bantuan sosial seperti Kartu Indonesia Pintar (KIP), Kartu Indonesia Sehat (KIS), hingga Program Keluarga Harapan (PKH).', 'Selain itu, sejak awal masa jabatannya, Joko Widodo juga mengupayakan reforma agraria dengan salah satunya melakukan percepatan penerbitan sertifikat hak atas tanah untuk mengurangi terjadinya sengketa lahan oleh karena ketiadaan sertifikat.\\nDi masa jabatannya yang kedua, Joko Widodo mengalihkan fokus pemerintahan pada pembangunan dan peningkatan kapasitas
```